

DAY 5- ASSEMBLE AND DISSEMBLE:

Tasks

1.

List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

Component	Short description
Motherboard	Main circuit board that connects the CPU, RAM, storage, expansion cards and other hardware.
CPU	Executes instructions and performs calculations for programs and the operating system.
CPU cooler	Removes heat from the CPU using a heatsink/fan or another cooling method.
RAM	Fast temporary memory used by active programs and data.
GPU	Processes graphics and video; may be integrated into the CPU or a separate dedicated card.
Storage drive	Stores the operating system, applications and files; commonly HDD, SATA SSD or NVMe SSD.
Power supply (PSU)	Converts AC power to regulated DC power for computer components.
Case fans	Move air through the case to help keep components within safe temperatures.
Expansion cards	Add capabilities such as Wi-Fi, sound, capture, networking or extra ports.
Cables	Carry power and data between the PSU, motherboard, storage and other components.

2.

Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.

Desktop PC Disassembly — Step-by-Step Procedure

Introduction

Disassembling a desktop PC involves carefully removing its components in a safe and organized order. Before starting, the computer must be completely powered off and disconnected from electricity. Photos should be taken at each stage so that the components and cable connections can be identified and the PC can be reassembled correctly later.

Step 1 : Shut Down and Unplug the PC

First, shut down the computer normally. Switch off the power supply if it has a PSU switch and disconnect the main power cable from the wall and the PC. Disconnect the keyboard, mouse, monitor and other peripherals.

Caption: The PC has been completely powered off and disconnected from the power source before beginning disassembly.

Photo/Diagram: Insert photo here.

Step 2 : Remove the Side Panel

Place the desktop cabinet on a stable surface. Remove the screws holding the side panel and carefully slide or lift the panel away from the cabinet.

Caption: The side panel has been removed to provide access to the internal components.

Photo/Diagram: Insert photo here.

Step 3 : Take a Photo of the Internal Layout

Before disconnecting anything, take a clear photograph of the inside of the PC. This provides a reference for where the cables and components were originally connected.

Caption: A photograph was taken before disassembly to record the original arrangement and cable connections.

Photo/Diagram: Insert photo here.

Step 4 : Disconnect the Power and Data Cables

Carefully disconnect the cables connected to the motherboard, storage drives, GPU, fans and other components. Hold connectors by their plastic ends rather than pulling the wires.

Caption: The power and data cables were carefully disconnected from the components.

Photo/Diagram: Insert photo here.

Step 5 : Remove the RAM

Locate the RAM modules on the motherboard. Push the retaining clips at the ends of the RAM slot outward. Hold the RAM module by its edges and gently remove it from the slot.

Caption: The RAM module was removed by releasing the slot clips and handling the module by its edges.

Photo/Diagram: Insert photo here.

Step 6 : Remove the HDD/SSD

Locate the storage drive. For a SATA HDD or SSD, disconnect the SATA data and power cables and remove the mounting screws or bracket. For an M.2 SSD, remove its retaining screw and gently slide the drive out of the slot.

Caption: The storage drive was disconnected and removed from its mounting position.

Photo/Diagram: Insert photo here.

Step 7 : Remove the Graphics Card

If a dedicated GPU is installed, disconnect its power cable if present. Remove the screw securing the graphics card to the case

and release the PCIe slot latch. Carefully lift the graphics card out of the PCIe slot.

Caption: The graphics card was removed from the PCIe expansion slot after disconnecting its power and releasing the retaining mechanism.

Photo/Diagram: Insert photo here.

Step 8 : Disconnect Remaining Motherboard Connections

Disconnect the remaining motherboard connections, including the 24-pin ATX power connector, CPU power connector, front-panel connectors, USB headers, audio connector and fan connectors.

Caption: The remaining motherboard cables and connectors were disconnected carefully before removing the motherboard.

Photo/Diagram: Insert photo here.

Step 9 : Remove the Motherboard

Check that all cables and components connected to the motherboard have been removed. Remove the screws securing the motherboard to the case. Hold the motherboard by its edges and carefully lift it out of the cabinet.

Caption: The motherboard was carefully removed from the cabinet after all connections and mounting screws were removed.

Photo/Diagram: Insert photo here.

Step 10: Final Inspection

After removing the motherboard, inspect the cabinet and remaining components. Keep all screws and components organized so they can be identified easily during reassembly.

Caption: The PC has been completely disassembled and the removed components have been organized for reassembly.

Photo/Diagram: Insert photo here.

Safety Precautions

- Always disconnect the PC from the power supply before opening the cabinet.
- Use an anti-static wrist strap if available.
- Handle RAM, the motherboard and expansion cards by their edges.
- Never pull cables by the wires.
- Do not force connectors, screws or components.
- Keep screws in a small container so they are not lost.
- Work on a clean, dry and stable surface.
- Take photographs before disconnecting cables.
- Follow the motherboard or computer manufacturer's manual when necessary.

Conclusion

The desktop PC was safely disassembled by following a systematic sequence: disconnecting power, removing the side panel, documenting the internal layout, disconnecting cables, removing RAM and storage, removing expansion cards and finally removing the motherboard. Taking photographs at each stage helps identify the components and makes the reassembly process easier and safer.

3.

Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

Desktop PC Reassembly — Step-by-Step Procedure

Introduction

After disassembling the desktop PC, the components must be installed and connected in the correct order. Reassembly should be done carefully to avoid damaging components or forgetting important connections. The PC should be tested before the cabinet is completely closed.

Step 1 — Install the Motherboard

Place the motherboard inside the cabinet and align it with the mounting holes and rear I/O opening. Secure it using the correct screws.

Caption: The motherboard was placed and secured inside the cabinet using the motherboard mounting screws.

Why this order matters: The motherboard is the main platform for the other components, so it should be installed before connecting most internal components.

Photo/Diagram: Insert photo here.

Step 2 — Install the RAM

Align the notch on the RAM module with the notch in the motherboard's RAM slot. Press the module down evenly until the retaining clips lock into place.

Caption: The RAM module was installed into the appropriate DIMM slot and secured by the retaining clips.

Why this order matters: Installing RAM before routing cables makes the slots easier to access and reduces the chance of damaging nearby components.

Photo/Diagram: Insert photo here.

Step 3 — Install the Storage Drive

Install the HDD/SSD in its mounting location. For a SATA drive, connect the SATA data and power cables. For an M.2/NVMe SSD, insert it into the M.2 slot at an angle and secure it with the retaining screw.

Caption: The storage drive was installed and connected to the motherboard and power supply as required.

Why this order matters: Storage should be securely mounted and connected before cable management and final assembly.

Photo/Diagram: Insert photo here.

Step 4 — Install the Graphics Card

Insert the GPU into the appropriate PCIe slot. Press it down gently until the slot latch engages, secure it to the case with the required screw and connect its power cable if the GPU requires one.

Caption: The graphics card was installed into the PCIe slot and secured to the cabinet.

Why this order matters: The GPU needs unobstructed access to the PCIe slot and should be installed before closing the case.

Photo/Diagram: Insert photo here.

Step 5 — Connect the Motherboard Power Cables

Connect the main 24-pin ATX power connector to the motherboard. Then connect the CPU power connector near the processor.

Caption: The motherboard and CPU power connectors were connected to supply power to the motherboard and processor.

Why this order matters: These connections provide essential power required for the computer to start.

Photo/Diagram: Insert photo here.

Step 6 — Connect Storage and GPU Power

Connect the SATA power cable to a SATA HDD/SSD if applicable. If the GPU requires additional power, connect the appropriate PCIe power connector from the PSU.

Caption: The required storage and graphics-card power connections were restored.

Why this order matters: These components need their power connections before the system can operate correctly.

Photo/Diagram: Insert photo here.

Step 7 — Connect Front-Panel and Other Headers

Reconnect the case's power switch, reset switch, LED connectors, front USB, front audio and other motherboard headers according to the motherboard manual.

Caption: The front-panel and case connectors were reconnected to their appropriate motherboard headers.

Why this order matters: These connections allow the case buttons, LEDs and front ports to function correctly.

Photo/Diagram: Insert photo here.

Step 8 — Connect Case Fans

Reconnect the case fans to the appropriate motherboard fan headers or fan controller. Make sure the fan blades can rotate freely.

Caption: The case fans were connected to provide proper airflow through the cabinet.

Why this order matters: Proper cooling is necessary before running the computer under load.

Photo/Diagram: Insert photo here.

Step 9 — Perform a Final Inspection

Before switching on the PC, check that:

- RAM is fully seated.
- GPU is fully seated.
- Storage drive is secure.
- 24-pin and CPU power connectors are connected.
- GPU power is connected if required.
- Front-panel connectors are connected correctly.
- Fans are connected and unobstructed.
- No loose screws or tools are left inside the cabinet.
- No cables are touching the fans.

Caption: A final inspection was performed before powering on the reassembled PC.

Photo/Diagram: Insert photo here.

Step 10 — Power On and Test the PC

Reconnect the power cable and switch on the PSU if it has a switch. Start the computer and check whether it reaches the BIOS/UEFI screen and then boots into the operating system.

Verify that:

- The correct amount of RAM is detected.

- The storage drive is recognized.
- The GPU is detected.
- The keyboard and mouse work.
- The case and CPU fans are operating normally.

Caption: The reassembled computer was powered on and checked to confirm that all major components were functioning correctly.

Photo/Diagram: Insert photo or screenshot here.

Why the Order Matters

The order of reassembly is important because some components can block access to others. Installing the motherboard first provides the base for the remaining components. RAM and storage are easier to install before cables become crowded, while power and front-panel cables are connected after the main components are securely installed. Performing a final inspection before powering on helps prevent loose connections, damaged components and short circuits.

Conclusion

The desktop PC was successfully reassembled by reinstalling the motherboard, RAM, storage, GPU and other components, followed by reconnecting the required power, data, front-panel and cooling cables. The system was then powered on and checked to verify that the hardware was correctly recognized and the operating system booted normally.

4.

Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

✓	Safety precaution / tool	Why it matters
■	Power cable disconnected	Prevents electrical hazards and accidental startup.
■	Anti-static wrist strap / ESD precautions	Reduces risk of electrostatic damage to components.
■	Correct screwdriver set	Prevents stripped screws and makes removal safer.
■	Clean, stable workspace	Prevents dropped parts, dust contamination and accidental damage.
■	Small container for screws	Keeps screws organized and prevents losing them.
■	Device/motherboard manual	Provides correct slot locations, cable headers and mounting instructions.
■	Good lighting	Makes small connectors, labels and screws easier to see.
■	Phone/camera	Useful for documenting cable positions before disassembly.
■	Do not force connectors	Prevents broken clips, pins and ports.
■	Handle components by edges	Reduces contact with sensitive contacts and circuitry.